

Generic external axis over Modbus TCP — protocol

One register map for any external axis a PLC fronts: a linear track carrying the robot (Pamet: Kinco PLC → Lichuan servo), a rotary positioner carrying the part, or any drive behind a PLC that can serve Modbus TCP. rw2 owns the axis through this map alone. The robot controller's own external-axis coupling is never used (owner ruling 2026-09-28/29). The map is the contract between the driver in this repository and the PLC program; both sides are written against this file.

Lineage: epic-065 external-axis-motion (Delta VFD-C2000 positioner, `delta-positioner` `DeltaRegisters.cs`, FR-11 dead-commander watchdog) and epic-065 feature-007 (Pamet track directly from the Kinco PLC).

Transport

Item	Value
Transport	Modbus TCP, port 502 (configurable)
Unit id	1 (configurable)
Register type	Holding registers only (FC03 read, FC06/FC16 write). No coils: one register file is simpler to map in every PLC.
Block placement	Two blocks. Command block at base <code>C</code> (default <code>0</code>), status block at base <code>S</code> (default <code>100</code>). Offsets inside a block are fixed; only the two bases are configurable, on both sides, for a PLC whose register file cannot start at 0/100. The addresses below are written for the defaults.
Word order for 32-bit values	low word first (register <code>n</code> = bits 0–15, register <code>n+1</code> = bits 16–31), two's complement; each 16-bit register big-endian on the wire (Modbus standard)
Position unit	0.001 mm for a linear axis, 0.001° for a rotary axis (the driver's configured kind decides; the map is the same)
Velocity unit	0.001 unit/s; acceleration 0.001 unit/s ²
Consistency	The PLC answers one FC03 read of the whole status block (15 registers) from one scan's image , and applies one FC16 write before or after a scan, never in the middle of one.
Map version	register 114 = 1 for this document

Command block — written by the driver, read by the PLC (holding registers `C+0 ... C+11`)

Addr	Name	Type	Meaning
0	Command	bitfield	bit 0 Enable (level: 1 = servo on). bit 1 Home , bit 2 MoveAbsolute , bit 3 MoveVelocity , bit 4 Stop , bit 5 Reset — edge-triggered: the PLC acts on 0→1 and the driver clears the bit after <code>CommandAck</code> echoes <code>CommandSeq</code> . Stop has priority over every other bit in the same word. Bits 6–15 are 0.

Addr	Name	Type	Meaning
1	CommandSeq	uint16	Incremented by the driver with every command write — a write that sets an edge bit or changes Enable (never 0; 65535 wraps to 1). The write that only clears edge bits after an ack does not increment it.
2–3	TargetPosition	int32	Target for MoveAbsolute, in position units.
4–5	Velocity	int32	Speed for MoveAbsolute (magnitude) and MoveVelocity (signed: sign = direction).
6–7	Acceleration	int32	Ramp for the next move; 0 = PLC default.
8	Heartbeat	uint16	FR-11: the driver writes an incrementing value (1...65535, never 0) at ≥ 5 Hz for the whole connection lifetime. The driver's default is 10 Hz.
9	LeaseOwner	uint16	FR-11 advisory lease: 0 = unowned; else the station-unique owner id of the driver holding the axis.
10	WatchdogFault	uint16	0 healthy; 1 = the PLC tripped the watchdog. Cleared only by the driver writing 0. Written by the PLC on trip.
11	WatchdogTrips	uint16	Trip counter since PLC power-up. PLC-owned, read-only for the driver.

Status block — written by the PLC, read by the driver (holding registers S+0 ... S+14)

Addr	Name	Type	Meaning
100	State	uint16	0 Disabled · 1 Standstill · 2 Homing · 3 DiscreteMotion · 4 ContinuousMotion · 5 (reserved) · 6 Stopping · 7 ErrorStop — the numbers of the SDK <code>AxisState</code> enum. Any other value is read by the driver as ErrorStop.
101	Flags	bitfield	bit 0 Homed · bit 1 InPosition · bit 2 LimitMin · bit 3 LimitMax · bit 4 HomeSensor · bit 5 DriveReady · bit 6 Moving
102–103	ActualPosition	int32	Position units. Valid only while <code>Homed</code> is set (an absolute encoder keeps it across power cycles; the map does not care which).
104–105	ActualVelocity	int32	Signed, 0.001 unit/s.
106	FaultCode	uint16	0 none · 1 drive fault · 2 limit switch tripped · 3 following error · 4 watchdog · 5 homing failed · 6 communication to drive lost · 7 safety stop (E-stop chain, guard) · 100+ vendor-specific (documented per PLC).
107	CommandAck	uint16	Echo of the last accepted <code>CommandSeq</code> .
108–109	TravelMin	int32	Machine limit, published by the PLC. The driver refuses targets outside <code>TravelMin..TravelMax</code> ; it never clamps.
110–111	TravelMax	int32	
112–113	MaxVelocity	int32	Machine limit, published by the PLC. The driver refuses faster speeds.
114	MapVersion	uint16	1. The driver reads the status block before its first write and refuses to attach to any other value, writing nothing.

Limits come from the machine through this block. If a PLC cannot publish them (`TravelMin` , `TravelMax` and `MaxVelocity` all zero), the driver uses the values from its own configuration and logs that it did; it never invents a number. When they are published they must satisfy `TravelMin < TravelMax` and `MaxVelocity > 0`; any other combination (a partial publication included) makes the driver refuse to attach.

Command semantics

- **Handshake.** Before the command write the driver writes the parameters it needs (`TargetPosition` , `Velocity` , `Acceleration` , registers 2–7) in one FC16, then `Command` and `CommandSeq` (registers 0–1) in a second FC16 — so the PLC never sees an edge before its parameters. The PLC accepts a command word whose `CommandSeq` differs from `CommandAck` and copies `CommandSeq` into `CommandAck` **in the same scan in which it enters the resulting state** (Homing, DiscreteMotion, ContinuousMotion, Stopping, Disabled, Standstill). A status read that shows the new ack therefore also shows the command's state, never the state from before it.
- **Acknowledge:** a command is accepted when `CommandAck == CommandSeq` . The driver waits ≤ 500 ms for the ack; no ack $\rightarrow$ SDK `CommunicationLost` for the command (the heartbeat keeps running); the driver clears the edge bit it set.
- **Sequence at attach:** the driver reads `CommandAck` and continues from `CommandAck + 1` , so a new driver never issues a sequence number the PLC already acknowledged.
- **Enable** (level): `Command.bit0 = 1` $\rightarrow$ PLC energises the drive; State leaves Disabled for Standstill. `bit0 = 0` $\rightarrow$ drive off, State = Disabled. After a watchdog trip or any entry into ErrorStop the PLC ignores bit 0 until a Reset, and after the Reset it energises only on a fresh 0 $\rightarrow$ 1 of bit 0 — an axis never re-energises by itself. A move with Enable low is ignored and answered by `FaultCode = 0` , State = Disabled ; the driver rejects it before writing (SDK `MotionError.Busy`).
- **Home** (edge): PLC runs its own homing sequence; State = Homing until done, then Standstill with `Homed` . The driver treats Home as callable from Disabled (it is the energise act: the driver sets Enable first). Failure $\rightarrow$ ErrorStop, `FaultCode = 5` .
- **MoveAbsolute** (edge): requires `Homed` ; PLC moves to `TargetPosition` at `Velocity` ; State = DiscreteMotion, then Standstill with `InPosition` . Target outside travel $\rightarrow$ the driver refuses (SDK `OutOfRange`) and writes nothing.
- **MoveVelocity** (edge): continuous motion at signed `Velocity` until Stop or a limit; State = ContinuousMotion. While `Homed` , reaching `TravelMin` / `TravelMax` is a controlled stop to Standstill (not a fault). A limit switch is always a fault: ErrorStop, `FaultCode = 2` .
- **Stop** (edge, priority): accepted in every state, Homing included; decelerate and hold; State = Stopping $\rightarrow$ Standstill. From Disabled or Standstill it is acknowledged and changes nothing. The driver sends Stop on the priority lane (see FR-11) and expects motion to cease within 200 ms of the write.
- **Reset** (edge): clears `FaultCode` and leaves ErrorStop $\rightarrow$ Disabled (Enable must be re-asserted, see Enable). Does not clear `WatchdogFault` (register 10 is cleared explicitly) and does not touch `Homed` . From any state other than ErrorStop it is acknowledged and changes nothing.
- **Limit switch recovery:** after a Reset with a limit switch still active, the PLC accepts motion only in the direction away from that switch.

FR-11 — dead-commander watchdog and advisory lease (verbatim from epic-065)

Motion must cease when the commander **dies**, not only when it calls Stop. Modbus TCP has no liveness semantics; a dead commander is indistinguishable from an idle one. Therefore:

- The driver runs a **connection-lifetime heartbeat** per axis, started at connect, stopped deliberately at disconnect, independent of motion, writing `Heartbeat` at ≥ 5 Hz. Heartbeat traffic shares the channel with move traffic; heartbeat deferral is **bounded at 200 ms** so a long move cannot starve the beat.
- The PLC **arms** the watchdog on the **first change** of `Heartbeat` it observes while `WatchdogFault = 0` and `LeaseOwner  $\neq$  0` (so an idle PLC is not faulted one second after boot), then **trips** when `Heartbeat` has not changed for **1 s**: Enable dropped, velocity zeroed, `State = ErrorStop`, `FaultCode = 4`, `WatchdogFault = 1`, `WatchdogTrips += 1`. A trip does **not** touch `Homed` (recovery = Reset + re-command, no re-home) and re-asserts the limit functions. After a trip the network disarms until `WatchdogFault` has been written 0 and a new beat arrives.
- **Clean release disarms.** When `LeaseOwner` is written 0 the watchdog disarms without tripping. A clean disconnect is: Stop $\rightarrow$ Enable 0 $\rightarrow$ stop beating $\rightarrow$ `LeaseOwner = 0`. `WatchdogTrips` therefore counts dead commanders only. A killed process cannot write 0, so the kill case is untouched.
- **Advisory lease:** before attaching, the driver reads `LeaseOwner`. 0 $\rightarrow$ take it (write own id). Own id $\rightarrow$ reattach. Foreign id $\rightarrow$ watch `Heartbeat` at the heartbeat interval; unchanged for one expiry window (1 s) $\rightarrow$ the lease has expired, take it; still changing $\rightarrow$ refuse (SDK `LeaseHeld`), keep watching until the configured lease timeout. Watching continuously rather than in discrete 1 s samples lets a successor attach ≤ 1.1 s after the incumbent's last beat. "Advisory" is stated plainly: Modbus has no compare-and-swap, so two drivers starting in the same window can both pass; the watchdog bounds the consequence. On a clean disconnect the driver writes `LeaseOwner = 0` if it still holds it.
- **At attach** the driver writes `WatchdogFault = 0` after winning the lease: a latched trip then belongs to a dead predecessor, and leaving it latched would leave the network disarmed for the new commander. `FaultCode = 4` and `ErrorStop` stay until a Reset, so the trip remains visible and distinguishable from a drive fault.
- Owner id is a station-unique non-zero 16-bit value from host configuration.
- Each heartbeat tick is: write `Heartbeat`, read registers 10–11, read the status block. That tick is the status cadence of the axis, moving or idle.
- Stop and cancellation take a **priority lane** on the channel that preempts queued move traffic (≤ 200 ms to a halted axis).

What the PLC program must do (checklist for the ladder / structured-text author)

1. Serve holding registers C+0...C+11 and S+0...S+14 on the configured unit id and port; refuse nothing, ignore writes to PLC-owned registers (11, 100–114).
2. Publish `MapVersion = 1`, `TravelMin/Max`, `MaxVelocity` from the axis parameters at start-up.
3. Mirror the drive: `State`, `Flags`, `ActualPosition`, `ActualVelocity`, `FaultCode` every PLC cycle; serve a status-block read from one scan's image.
4. Execute the command semantics above; echo `CommandSeq` into `CommandAck` in the scan that enters the command's state.
5. Implement the watchdog exactly as FR-11 states (arm on first change with a lease held, 1 s stall, trip actions, disarm until cleared, disarm on `LeaseOwner = 0`).
6. Pass the conformance checks below, CHK-01...CHK-16 (motion checks at commissioning, with an operator present).
7. Keep the project in a git repository next to this file's copy (epic-065 risk R-1: an unversioned PLC program is a drive that answers every register and moves nothing after replacement).

Scope of map version 1

- One axis per block pair. A PLC serving two axes places a second block pair at other bases and is configured as two devices.
- A rotary axis has limited travel: `ActualPosition` is absolute and never wraps. Endless rotation is not in version 1.

Conformance checks

This checklist is executable. Two tools run it verbatim against a PLC, with no `rw2` involved:

- the C# test app, `dotnet ModelingEvolution.GenericAxis.TestApp.dll --check <host>[:port] ...`;
- the Python tool, `python -m generic_axis_check <host>[:port] ...` (in `python/generic_axis_check/`).

The simulator must pass the full list, motion included, in CI. The table below is the single source of truth. A check id missing from either tool, or present in a tool but not here, is a defect, and each tool has a test that compares its ids with this table.

Command line (identical in both tools)

Argument	Default	Meaning
<code><host>[:port]</code>	port 502	The PLC.
<code>--unit N</code>	1	Unit id.
<code>--command-base N</code> , <code>--status-base N</code>	0, 100	Block bases.
<code>--owner-id N</code>	65535	The checker's lease id. CHK-11 uses 65534 as the foreign id. Ids 65534–65535 are reserved for conformance tools; stations never use them.
<code>--allow-motion</code>	off	Runs CHK-12...CHK-16. Without it they are SKIPPED. Only with an operator at the machine and the travel clear.
<code>--tolerance X</code>	0.1	Position check threshold in axis units (CHK-13). This is a checker threshold, not a machine number.
<code>--report PATH</code>	none	<code>*.md</code> : writes the Markdown report there and the JSON report next to it as <code>*.json</code> . <code>*.json</code> : writes the JSON report only. Any other extension is a usage error (exit 2). The Markdown report always goes to stdout.

Exit codes: 0 = no FAIL (SKIPPED allowed) · 1 = at least one FAIL · 2 = usage error · 3 = refused to start (the axis is held by a live foreign lease, meaning another commander such as `rw2` is attached; stop it first).

Rules for every run

- **Pre-flight.** Read `LeaseOwner` and watch `Heartbeat` for 1 s. If a foreign owner is beating, exit 3, write nothing, and report every check SKIPPED.
- **Order.** Checks run in id order. A check whose prerequisite FAILED or was SKIPPED is SKIPPED, and its message names the prerequisite.
- **Timing.** Timing checks poll the status block every **20 ms**. Every duration is measured from the completion of the triggering write to the first read that shows the effect, and is reported in ms.
- **Units.** All positions and velocities are raw register values ÷ 1000, in the PLC's axis unit (mm or °).

- **Cleanup, always, even after a FAIL or Ctrl-C:** Stop edge if State is 2, 3 or 4 · clear edge bits · Enable 0 · `WatchdogFault = 0` if the checker caused a trip · `LeaseOwner = 0` if it holds the checker's id. Each cleanup write is logged in the report.
- **Lease and beat between checks.** From CHK-06 onwards the checker holds the lease under its own id and beats, except where a check says it stops.
- **Each check restores.** Every check ends with the axis in State 0 or 1, no latched fault, the lease held and the beat running: Reset, `WatchdogFault = 0` and re-take as needed. If it cannot restore, it FAILs with the reason, and every later check is SKIPPED.
- **Beat.** Whenever a check says "beat", the checker writes `Heartbeat` every 100 ms (1...65535, never 0) from its own loop, not through a driver.

The checks

Id	Title	Protocol section	Needs	Procedure	PASS when
CHK-01	Transport and unit	Transport	—	TCP connect (2 s timeout), then FC03 of S+0...S+14 on the unit.	Connected, and the read answers with no Modbus exception.
CHK-02	Map version	Status block	01	Read S+14.	<code>MapVersion == 1</code> .
CHK-03	Machine limits published	Status block, "Limits come from the machine"	02	Read S+8...S+13.	Not all zero, <code>TravelMin < TravelMax</code> , <code>MaxVelocity > 0</code> . The values are reported.
CHK-04	Status mirror cadence	Status block; FR-11 tick	02	30 status-block reads, one every 100 ms.	All 30 answer, the slowest round trip is ≤ 100 ms, and <code>State</code> $\in \{0,1,2,3,4,6,7\}$ in every read.
CHK-05	32-bit word order and driver ownership of parameters	Transport (word order); Command block	02	Write C+2...C+3 = <code>[0x0002, 0x0001]</code> (65 538), read back. Write -2 as <code>[0xFFFE, 0xFFFF]</code> , read back. Wait 1 s and read again. No edge bit is set.	Both values read back exactly, and are unchanged after 1 s (the PLC does not write driver-owned registers). The PLC's <i>interpretation</i> of the order is proven by CHK-03 (sane limits) and CHK-13 (it arrives where it was sent).
CHK-06	Enable handshake (level)	Command semantics: Handshake, Enable	02	Precondition State 0 or 1. Take the lease (checker id), beat. Write <code>[Enable, seq+1]</code> , then after the ack <code>[0, seq+2]</code> .	Each ack arrives in ≤ 500 ms. State is 1 within 5 s after Enable 1 and 0 within 5 s after Enable 0. Ack ms and state ms are reported. Energises the drive and commands no motion.
CHK-07	Reset handshake (edge)	Command semantics: Reset, Acknowledge	06	From State 0 write <code>[Reset, seq+1]</code> . After the ack, clear the edge <code>[0, seq+1]</code> .	Ack in ≤ 500 ms, <code>State</code> stays 0 and <code>FaultCode</code> stays 0 (Reset outside ErrorStop is a no-op).

Id	Title	Protocol section	Needs	Procedure	PASS when
CHK-08	Watchdog trips on a stalled beat	FR-11	06	Hold the lease, <code>WatchdogFault = 0</code> , beat for 2 s, then stop beating. Keep polling.	<code>WatchdogFault == 1</code> , <code>WatchdogTrips + 1</code> , <code>State == 7</code> , <code>FaultCode == 4</code> , all within 1.0–1.5 s of the last beat. The trip time is reported.
CHK-09	Watchdog disarms after a trip and re-arms on clear	FR-11	08	Without clearing, beat 1 s: no second trip is counted. Then Reset edge, <code>WatchdogFault = 0</code> , beat 2 s (must not trip), stop beating.	No trip while latched. No trip while beating. A second trip within 1.0–1.5 s with <code>WatchdogTrips + 1</code> . Recovery afterwards is Reset plus <code>WatchdogFault = 0</code> .
CHK-10	Clean release disarms	FR-11 "Clean release disarms"	08	Beat 2 s, write <code>LeaseOwner = 0</code> , stop beating, wait 2 s.	No trip: <code>WatchdogFault == 0</code> and the trip count is unchanged.
CHK-11	Advisory lease	FR-11 "Advisory lease"	02	(a) <code>LeaseOwner = 0</code> → the checker's lease client takes it and reads back its id, then releases. (b) Write <code>LeaseOwner = 65534</code> and beat as that incumbent. The checker's lease client, with a 3 s timeout, must refuse. (c) Stop the incumbent's beat while the client watches.	(a) Read-back equals the checker id. (b) Refused with LeaseHeld naming 65534 after 3 s, and <code>LeaseOwner</code> is still 65534. (c) Taken within 2 s of the incumbent's last beat, with the time reported. Any trip caused by (c) is cleaned up.
CHK-12	Home	Command semantics: Home	06, -- allow-motion	Enable, then Home edge.	Ack in ≤ 500 ms. Then <code>State == 1</code> with <code>Homed</code> within 120 s, and <code>FaultCode == 0</code> . Duration reported.
CHK-13	MoveAbsolute to TravelMin + 10	Command semantics: MoveAbsolute	03, 12	Target <code>TravelMin + 10</code> , velocity 10 % of <code>MaxVelocity</code> , acceleration 0.	Ack with <code>State == 3</code> in ≤ 500 ms. Then <code>State == 1</code> with <code>InPosition</code> , and <code>abs(ActualPosition - target) ≤ --tolerance</code> . The error and duration are reported.
CHK-14	Stop mid-move	Command semantics: Stop; FR-11 priority	13	MoveAbsolute toward <code>TravelMin + (TravelMax - TravelMin)/2</code> at 10 %. Once <code>abs(ActualVelocity) ≥ 90</code> % of the commanded speed (or after 2 s), write Stop.	Ack in ≤ 500 ms. <code>ActualVelocity == 0</code> and <code>State == 1</code> within 200 ms of the Stop write. The time is reported.

Id	Title	Protocol section	Needs	Procedure	PASS when
CHK-15	MoveVelocity	Command semantics: MoveVelocity	13	MoveVelocity at +1 % of MaxVelocity (away from TravelMin) for 1 s, then Stop.	Ack with State == 4, ActualVelocity > 0 during the run, then State == 1 after Stop within 200 ms.
CHK-16	Kill test	FR-11	15	MoveVelocity at +1 %, then stop beating. The connection stays open, and polling continues.	Trip (FaultCode 4, State 7) within 1.0–1.5 s of the last beat. ActualVelocity == 0 within 200 ms of the trip. Homed is still set. Both times are reported.

Report schema

JSON (schema: "generic-axis-conformance/1"). Both tools emit exactly these fields:

```
{
  "schema": "generic-axis-conformance/1",
  "mapVersion": 1,
  "tool": { "name": "generic-axis-check", "language": "python", "version": "1.0.0"
},
  "target": { "host": "192.168.58.20", "port": 502, "unit": 1, "commandBase": 0,
"statusBase": 100 },
  "allowMotion": false,
  "startedAt": "2026-09-29T10:15:02Z",
  "finishedAt": "2026-09-29T10:15:31Z",
  "summary": { "result": "PASS", "pass": 11, "fail": 0, "skipped": 5 },
  "checks": [
    { "id": "CHK-08", "title": "Watchdog trips on a stalled beat", "section": "FR-
11",
      "result": "PASS", "durationMs": 3140,
      "message": "trip after 1.12 s",
      "observed": { "tripAfterMs": 1120, "watchdogTrips": 3, "faultCode": 4,
"state": 7 } },
    { "id": "CHK-12", "title": "Home", "section": "Command semantics: Home",
      "result": "SKIPPED", "durationMs": 0, "message": "needs --allow-motion",
"observed": {} }
  ],
  "cleanup": [ "C+0 = 0x0000 (Enable 0)", "C+9 = 0 (release lease)" ]
}
```

- result is PASS, FAIL or SKIPPED. summary.result is FAIL if any check failed, otherwise PASS.
- observed keys are camelCase. Numbers are raw integers or durations in ms. language is python or csharp.

The Markdown report has three parts. A heading gives the tool, the target and the UTC time. A table follows with the columns Id | Title | Result | Observed | Protocol section. Last comes the cleanup list. The last line is RESULT: PASS or RESULT: FAIL.

Reference: the Delta positioner ladder this generalises

```
M0 -> M1025 (RUN) + M1040 (servo on)
M4 -> edges: MOV 0/1 D1060 (speed/position mode); FREQ D110 D111 D112
M5 -> M1026 (direction: OFF = forward, ON = reverse)
M6 + edge X7 -> DSUB D1051 D120 D122; DMOV D1051 D120    (home latch)
D130 heartbeat · D131 lease owner · D132 latched watchdog fault · D133 trip counter
```

(`delta-positioner/src/RocketWelder.SDK.Devices.Motion.Delta/DeltaRegisters.cs`, `epic-065`
`current-state.md`.)